

SANTIRAM ENGINEERING COLLEGE (AUTONOMUS)



TITLE : YOLO MODEL SAFETY HELMET DETECTION IN CONSTRUCTION SCENES WITH DEEP LEARNING

UNDER THE GUIDANCE OF

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ABSTRACT

Ensuring safety compliance on construction sites is critical to reducing workplace hazards. This project introduces an intelligent helmet detection system using the YOLOv8 (You Only Look Once, version 8) object detection algorithm to identify whether workers are wearing safety helmets in real-time. The system is designed for deployment in construction environments to enhance on-site safety monitoring through automated video surveillance. Leveraging the high accuracy and speed of YOLOv8, the model can effectively detect helmet presence in complex and dynamic scenes. The application is built using a Flask-based Python backend and a responsive front-end interface developed with HTML, CSS, and JavaScript. This lightweight and scalable solution offers real-time alerts and visual feedback, aiding construction managers in ensuring regulatory compliance and preventing accidents. The proposed system is suitable for integration with CCTV feeds or on-site monitoring setups, offering a cost-effective tool for occupational safety enforcement.

Keywords: Helmet Detection, YOLOv8, Construction Safety, Flask, Object Detection, Computer Vision.

OBJECTIVE OF PROJECT

The primary objective of this project is to develop an AI-powered helmet detection system using YOLOv8 that can accurately and efficiently identify the presence or absence of safety helmets on workers in construction environments.

The key goals include:

1. **Automated Helmet Detection:** Implement a deep learning-based object detection model capable of identifying helmets in real-time across varying lighting, backgrounds, and angles.
2. **Real-Time Surveillance Integration:** Enable seamless integration with existing CCTV or webcam feeds for live monitoring.
3. **Alert Mechanism:** Generate instant alerts or visual cues when non-compliance is detected to help site supervisors take timely action.
4. **Web Interface:** Design a user-friendly web application using Flask (Python), HTML, CSS, and JS that displays the detection output and alerts.

INTRODUCTION

- Construction is one of the most vital yet dangerous sectors of the economy, with workers frequently exposed to hazardous conditions. Safety helmets, a core component of personal protective equipment (PPE), are essential for preventing head injuries.
- However, ensuring consistent usage of helmets on-site remains a persistent challenge. Traditional monitoring methods, including manual supervision and periodic inspections, are often inadequate, especially on large or complex sites. This calls for an automated, real-time solution to monitor and enforce helmet compliance. With advancements in artificial intelligence and computer vision, real-time object detection models like YOLO (You Only Look Once) offer powerful tools for visual monitoring.
- This project employs YOLOv8—an improved, faster, and more accurate variant of the YOLO family—to build an AI-based safety helmet detection system. The backend is implemented using Python and Flask, with a modern web front-end for visualization and control.
- The proposed system is capable of processing live video feeds to detect workers with or without helmets, providing immediate visual feedback and enabling site managers to take corrective action. The integration of such a system can dramatically reduce risks, improve regulatory compliance, and serve as a scalable safety solution across construction sites.

EXISTING SYSTEM

- Existing helmet detection systems primarily rely on manual inspection or outdated computer vision techniques. Traditional safety enforcement involves site supervisors monitoring workers visually or reviewing surveillance footage post-incident, both of which are prone to oversight and human error.
- Some existing digital solutions use motion sensors or color-based detection, which are unreliable in real-world construction environments due to changing lighting, varied helmet designs, and background noise.
- Other systems may use older versions of object detection algorithms like Haar cascades or classical CNNs, which lack the accuracy and speed needed for real-time deployment. Furthermore, many of these systems are not scalable, do not integrate well with existing site surveillance systems, and fail to provide immediate alerts or visual dashboards for action.
- This results in poor adoption, high maintenance costs, and insufficient safety enforcement. Most critically, existing solutions do not leverage the latest advancements in deep learning models such as YOLOv8, which provide superior performance in terms of detection accuracy, multi-object tracking, and low latency.
- There is a clear gap in the market for a modern, AI-driven, end-to-end helmet detection system that is both efficient and cost-effective.

DISADVANTAGES

- Relies heavily on manual monitoring, leading to human error.
- Outdated algorithms provide poor accuracy in complex scenes.
- Not suitable for real-time detection or large-scale deployment.
- Inability to handle diverse lighting and background variations.
- No instant alerting mechanism for non-compliance cases.
- Poor integration capabilities with existing CCTV infrastructure.
- Limited or no support for visual dashboards or user interfaces.
- High false positive/negative rates with traditional methods.

PROPOSED SYSTEM

- The proposed system is a real-time helmet detection application powered by YOLOv8, one of the latest and most accurate object detection models available. It leverages deep learning to automatically detect whether construction workers are wearing helmets during their work hours.
- The system processes live video streams using a Python-Flask backend and provides visual feedback via a responsive front-end built with HTML, CSS, and JavaScript. The model is trained on a dataset comprising diverse construction scenes with various lighting conditions, helmet types, and worker postures to ensure robustness and generalizability.
- When a worker is detected without a helmet, the system generates real-time visual alerts and can optionally trigger audio alarms or notifications. This application is designed to be lightweight, scalable, and compatible with existing surveillance systems, making it easy to deploy on any construction site.
- It can significantly reduce the burden on safety personnel by automating surveillance and ensuring strict helmet compliance. Additionally, the system architecture allows for future enhancements such as face recognition, report generation, and integration with centralized safety management dashboards.

ALGORITHM DEFINITION

What is YOLOv8?

YOLOv8 (You Only Look Once, version 8) is the latest iteration in the YOLO series of real-time object detection models, developed by Ultralytics. It is optimized for speed and accuracy, making it ideal for safety applications such as helmet detection in dynamic construction environments.

Unlike traditional two-stage detectors (like R-CNN), YOLO performs object classification and localization in a single forward pass, making it extremely fast and suitable for real-time surveillance systems.

YOLOv8 Architecture Overview

YOLOv8 includes the following components:

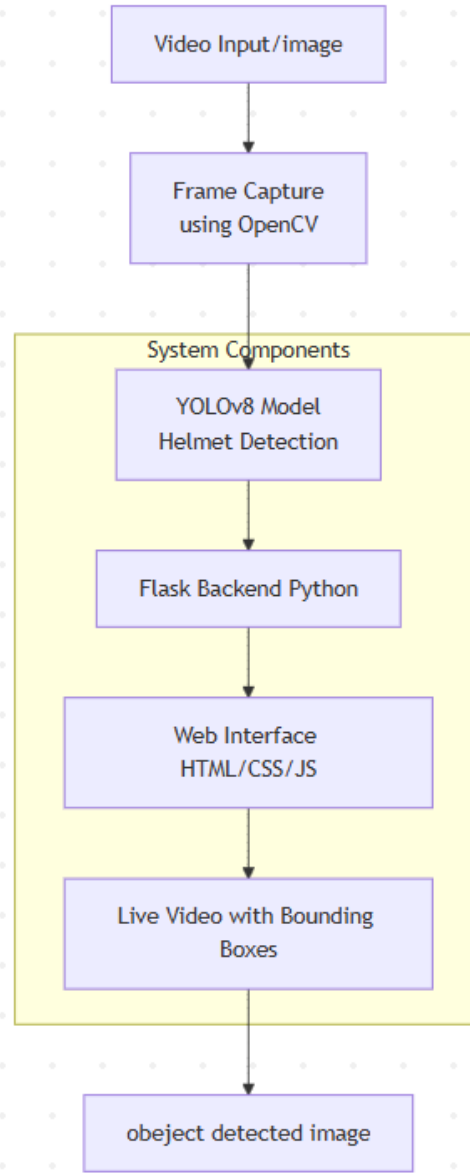
Backbone – CSP-Darknet-inspired feature extractor.

Neck – Path Aggregation Network (PAN) for multi-scale feature fusion.

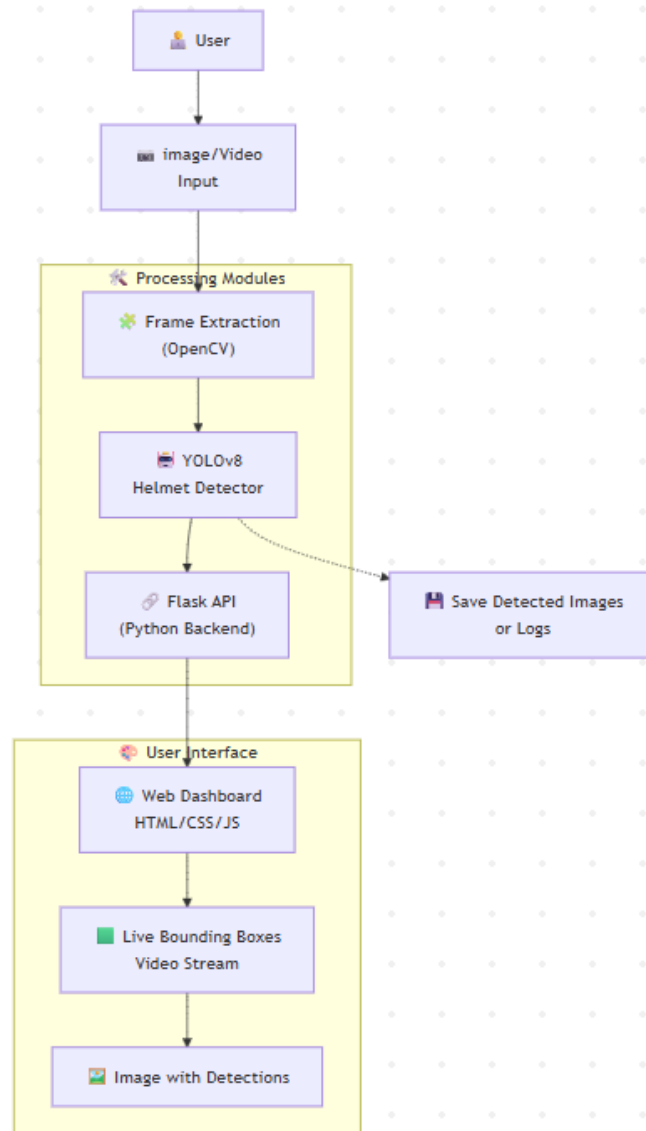
Head – Predicts bounding boxes, objectness, and class probabilities.

Anchor-Free Design – Unlike older YOLO versions, YOLOv8 supports anchor-free detection for faster convergence and simpler training

BLOCK DIAGRAM



ARCHITECTURE



Resource Requirements

Resources

01

Hardware

Processor: I5/Intel Processor, **RAM:** 8GB (min), **Hard Disk:** 128 GB, **Key Board:** Standard Windows Keyboard, **Mouse:** Two or Three Button Mouse)

02

Software

(**Operating System:** Windows 10, **Programming Language:** Python 3.10.8, **IDE:** VS Code)

03

Packages/Libraries

(Flask, Ultralytics , opencv, Smtplib, mysql-connector

MODULES

User Module

The User Module provides an interactive and user-friendly interface for safety officers or administrators to interact with the system.

Key Features:

- Login/Access Page: (Optional if authentication is used) Enables access to the monitoring dashboard.
- Live Video Feed Display: Shows the real-time camera or video stream with helmet detection overlays.
- Visual Alerts: When a person without a helmet is detected, the system displays a red bounding box or warning on-screen.
- Detection Status Panel: Shows real-time statistics such as total detections, helmet compliance rate, and alerts triggered.
- Responsive UI: Built using HTML, CSS, and JavaScript for cross-device compatibility and ease of use.
- View Logs (Optional): Allows users to review previous detection events and alert records (can be integrated later).

System Modules

The System Modules are responsible for processing the video feed, running the YOLOv8 model, managing backend logic, and sending responses to the frontend.

Key Components:

Video Capture Module:

- Accepts video input from CCTV, webcam, or IP camera.

- Converts video frames into a format usable by the detection module.

YOLOv8 Detection Engine:

- Processes each frame to detect presence or absence of helmets.

- Uses pre-trained or custom-trained YOLOv8 weights optimized for helmet detection in construction environments.

- Outputs class labels (Helmet / No Helmet) and bounding box coordinates.

Flask Backend API:

- Acts as the intermediary between the detection model and frontend.

- Routes: /upload, /video_feed, /alert handle different system tasks.

- Can optionally store logs or statistics in a database (e.g., SQLite).

ADVANTAGES

- Accurate detection using YOLOv8.
- Automated alerts for helmet non-compliance.
- High precision in varied construction environments.
- Scalable and integrable with existing CCTV systems.
- User-friendly web interface for easy monitoring.
- Reduces human effort and improves safety compliance.
- Lightweight backend with Python Flask.
- Future-ready architecture for extended PPE detection.

RESULTS

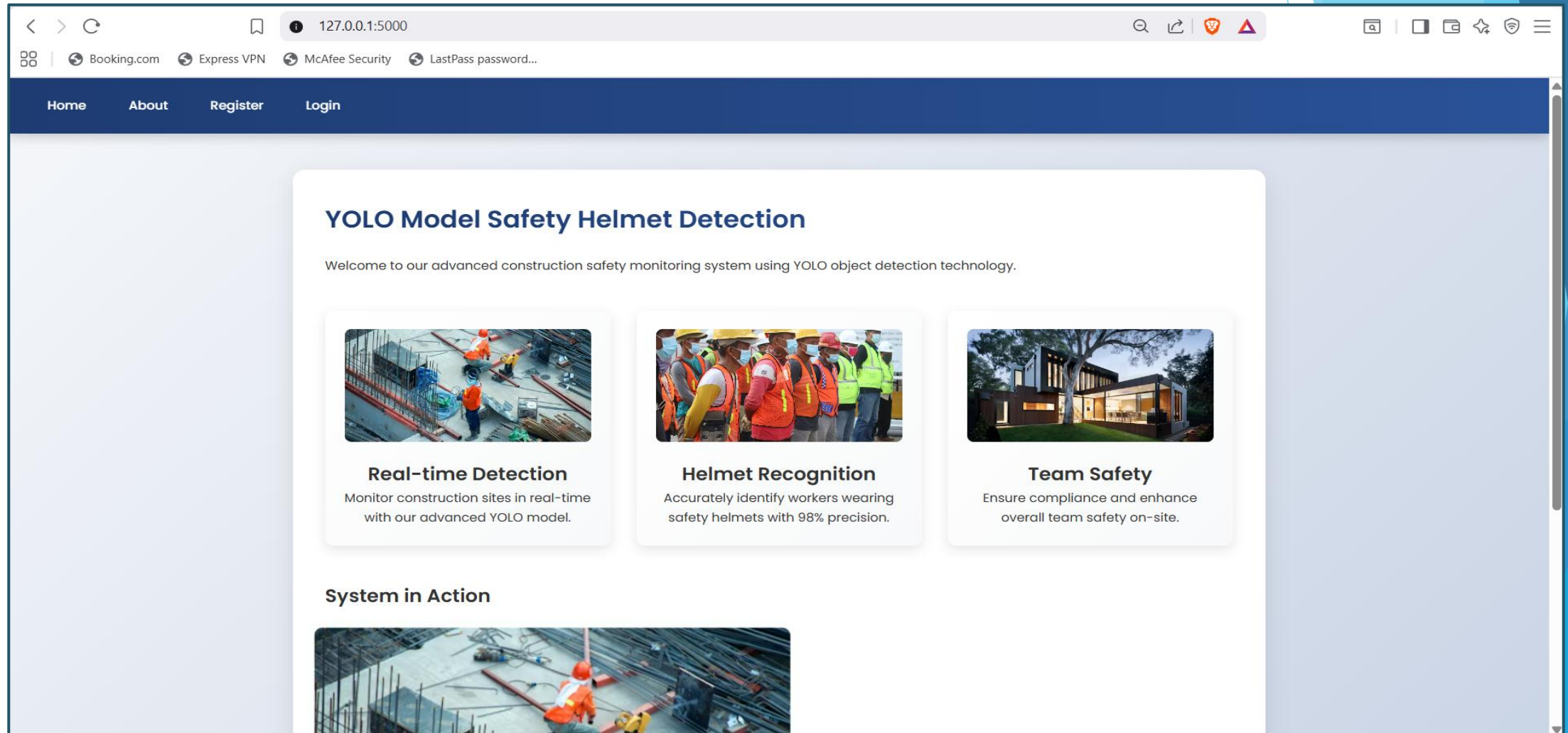


Fig : Home Page

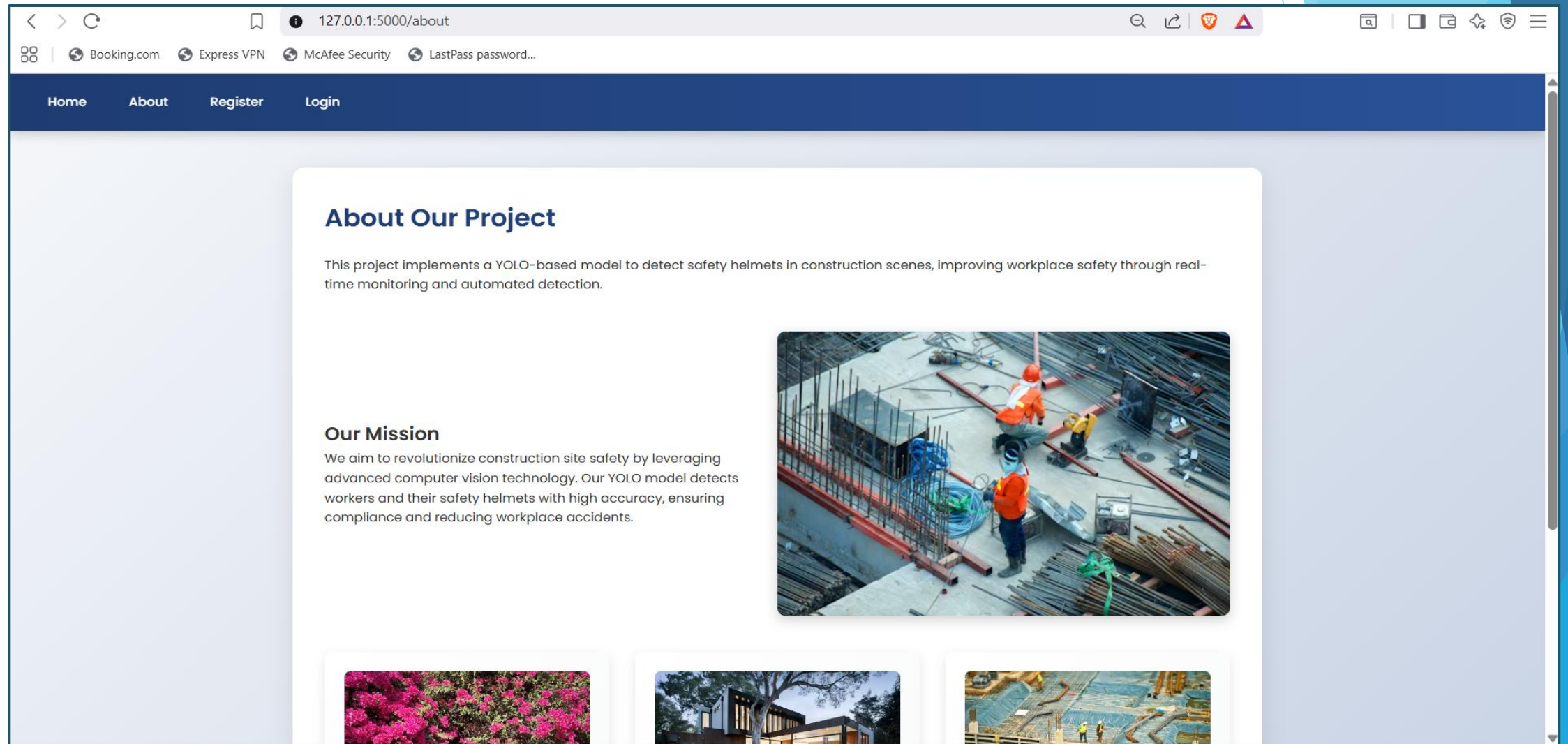
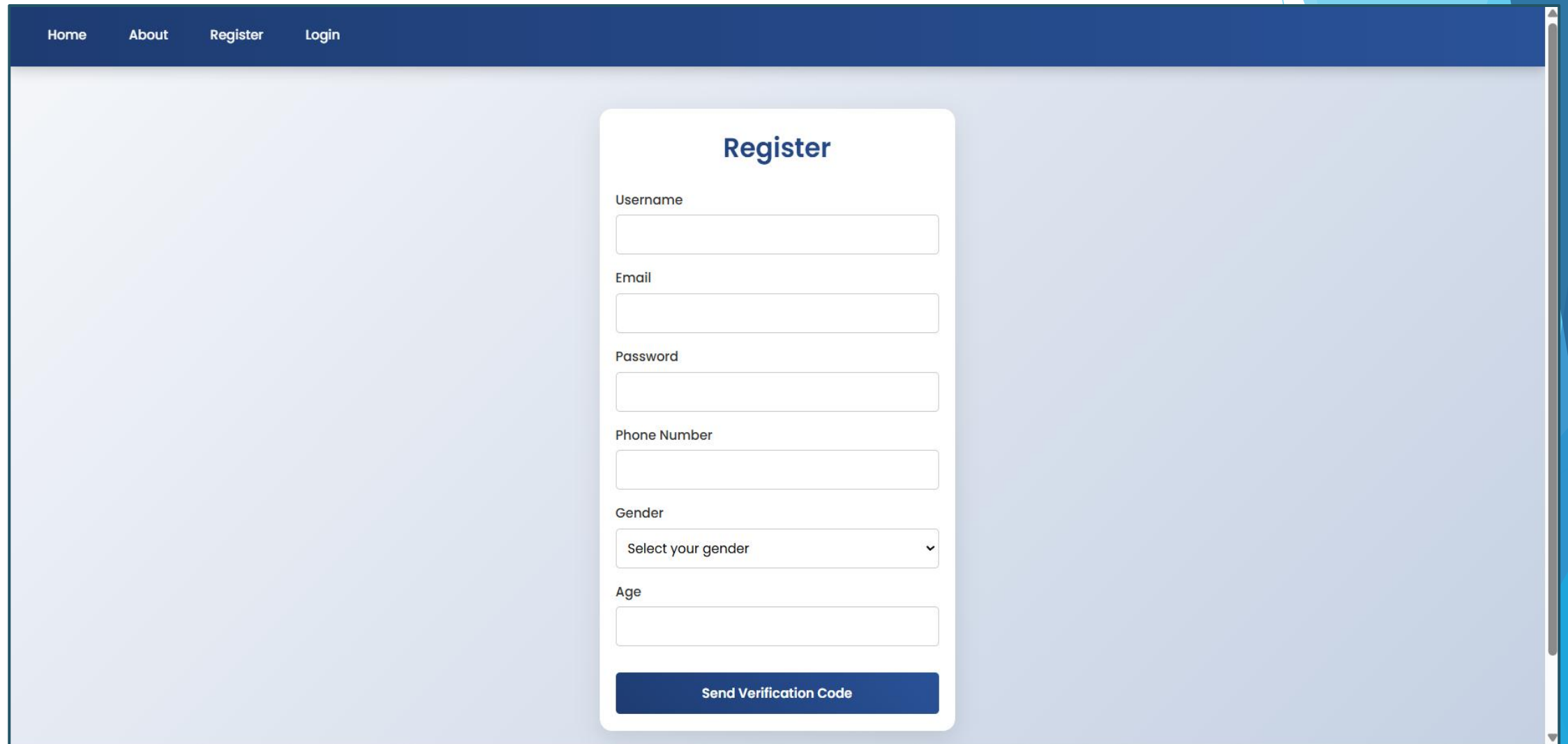


Fig : Home Page



The image shows a web browser window with a registration form. The browser's address bar is empty. The page has a dark blue header with navigation links: Home, About, Register, and Login. The 'Register' link is highlighted. The main content area has a light blue background. In the center, there is a white rounded rectangle containing the registration form. The form is titled 'Register' in a bold, dark blue font. It includes input fields for Username, Email, Password, and Phone Number. There is a dropdown menu for Gender with the text 'Select your gender' and a downward arrow. Below the dropdown is an input field for Age. At the bottom of the form is a dark blue button with the text 'Send Verification Code' in white. The browser window has a vertical scrollbar on the right side.

Home About Register Login

Register

Username

Email

Password

Phone Number

Gender

Select your gender ▼

Age

Send Verification Code

Fig : Registration Page

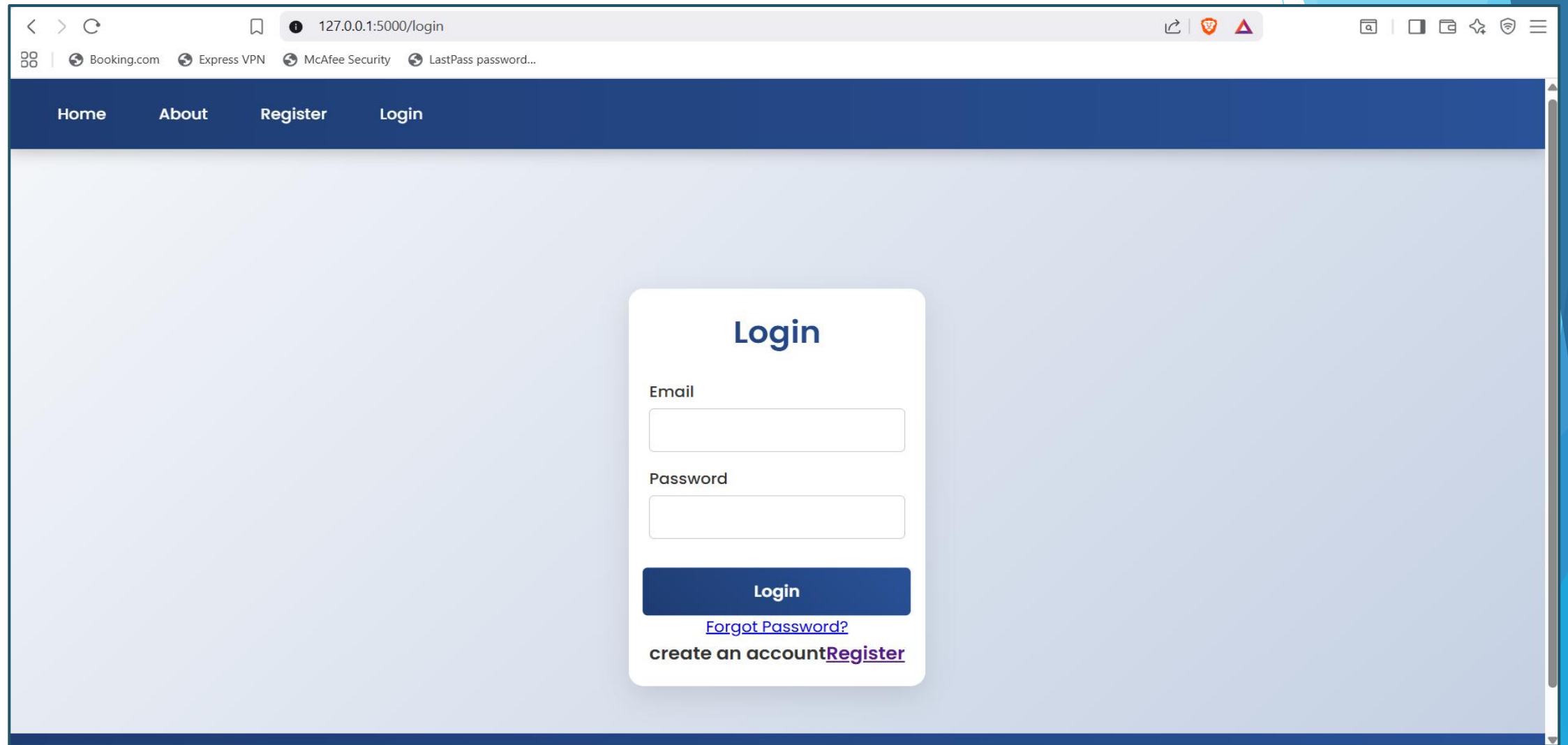


Fig : Login Page

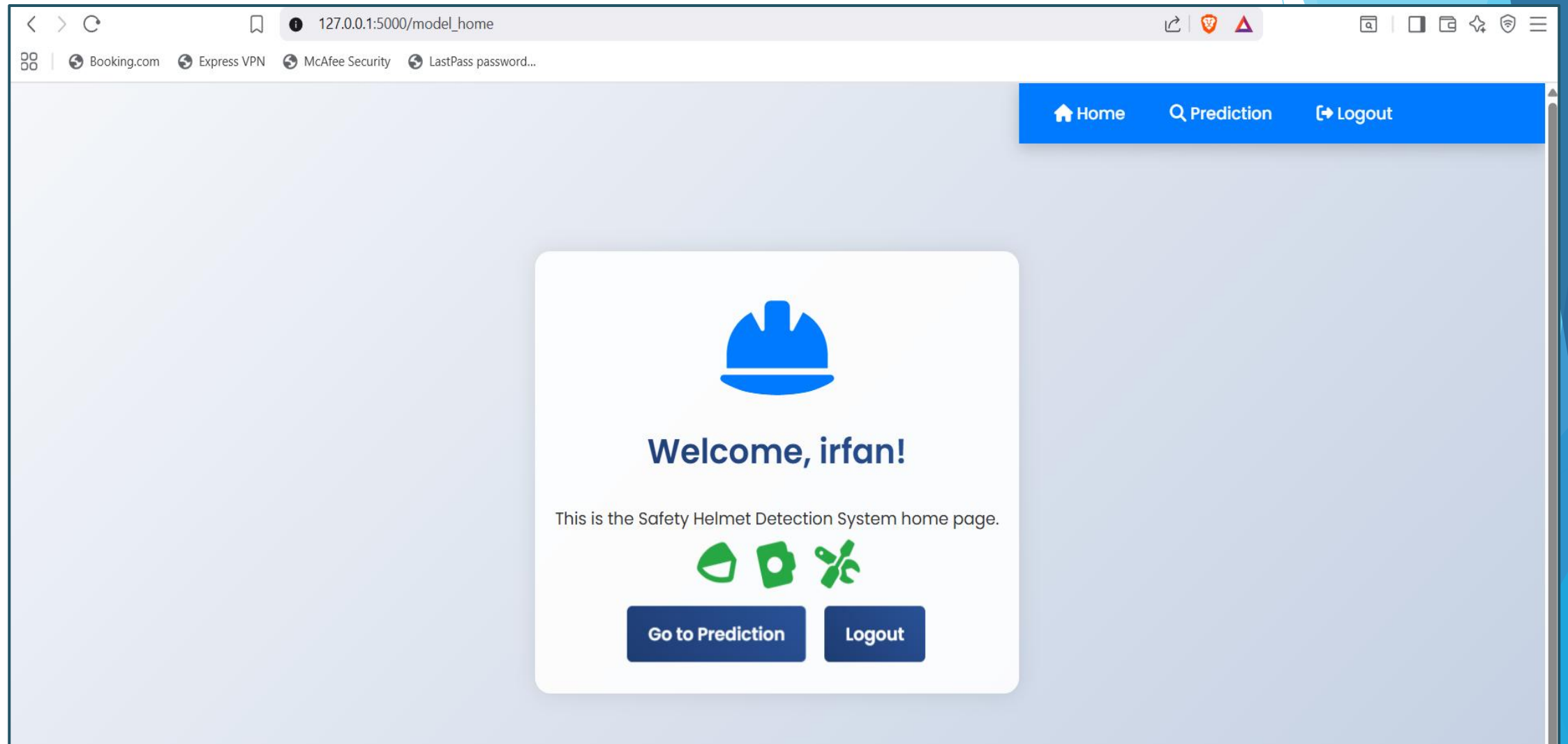


Fig : Prediction Page

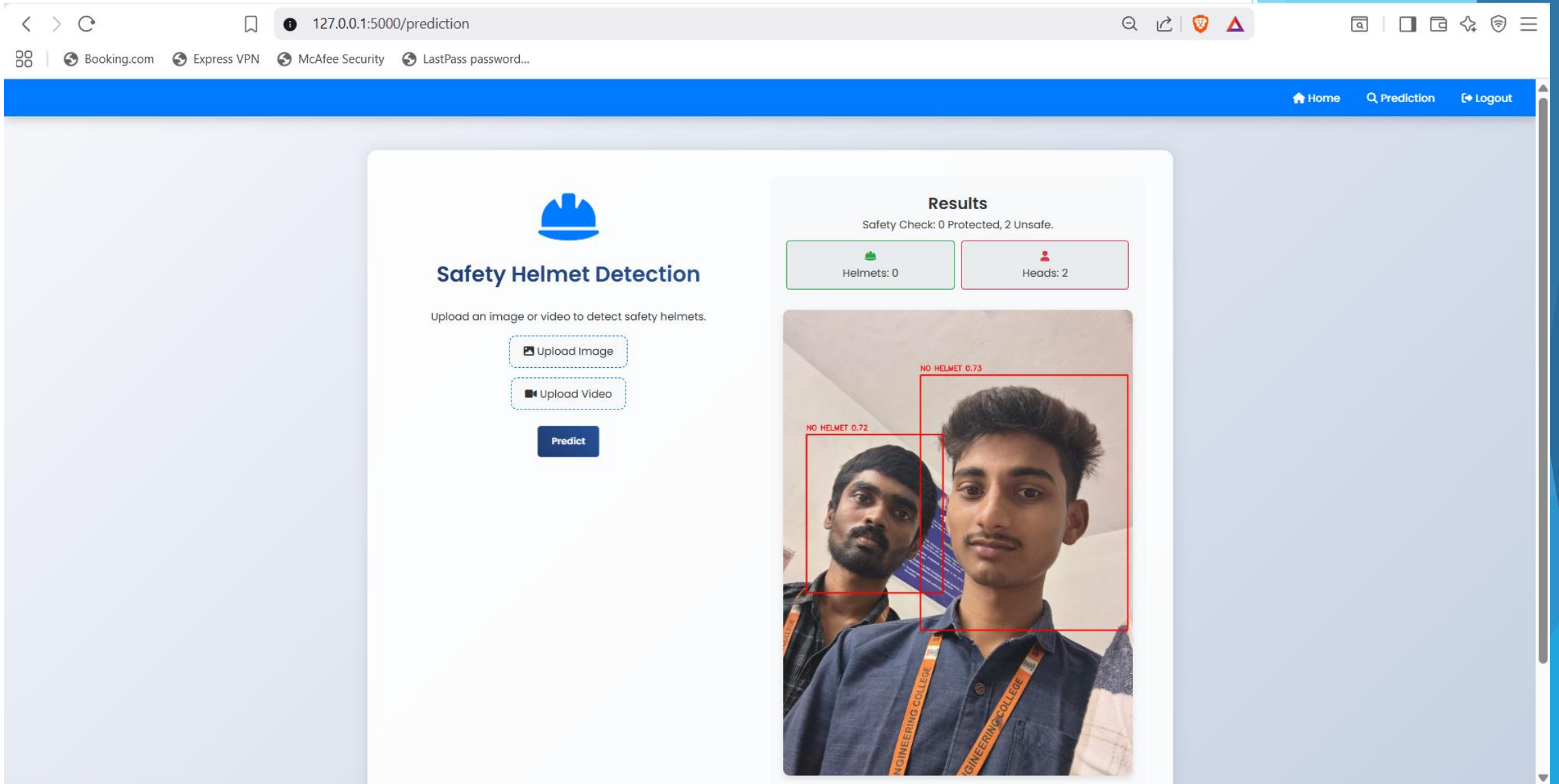


Fig : Output Page

CONCLUSION

- The proposed helmet detection system leveraging the YOLOv8 object detection algorithm presents a practical and highly effective solution for enhancing safety compliance in construction environments. By automating the process of identifying whether workers are wearing safety helmets, the system significantly reduces the dependence on manual monitoring, which is often error-prone and labor-intensive. YOLOv8, with its state-of-the-art architecture and real-time processing capabilities, ensures high accuracy and fast detection, even in complex, dynamic backgrounds typically found on construction sites.
- The integration of this model with a Flask-based backend and a responsive web interface using HTML, CSS, and JavaScript makes the solution lightweight, scalable, and user-friendly.

FUTURE SCOPE

- While the current system efficiently detects helmet usage in real-time using YOLOv8, there are several future enhancements that can elevate its functionality and scalability. One major enhancement would be the extension of the model to detect other personal protective equipment (PPE) such as safety vests, gloves, and goggles, creating a comprehensive worker safety monitoring system. Integrating face recognition can help in identifying workers individually, enabling automatic attendance tracking and personalized compliance logs. Another promising direction is the deployment of the system on edge devices like NVIDIA Jetson Nano or Raspberry Pi for real-time, low-power on-site monitoring, especially in remote areas without stable internet access.